

1

POLYMORPHS AND NEW PATH TO SYNTHESIZE TAFAMIDIS

CROSS-REFERENCE TO RELATED APPLICATIONS

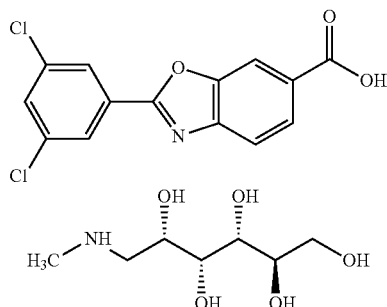
This application is a U.S. National Phase Application under 35 U.S.C. 371 of International Application No. PCT/EP2019/056317, filed on Mar. 13, 2019, which claims the benefit of Great Britain Patent Application No. 1804005.5, filed on Mar. 13, 2018. The entire disclosures of the above applications are incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates to a new path to synthesize crystalline Tafamidis polymorphs starting from a Tafamidis acetic acid adduct. In addition, this invention relates to a process to synthesize Tafamidis (2-(3,5-dichlorophenyl)-1,3-benzoxazole-6-carboxylic acid * (2R,3R,4R,5S)-6-(methyl-amino)-hexane-1,2,3,4,5-pentol), new Tafamidis crystalline polymorphs, a pharmaceutical compositions comprising the new crystalline polymorphs and to the therapeutic use of such new polymorphs.

BACKGROUND

Tafamidis is a drug used for the treatment of transthyretin-related hereditary amyloidosis.



Transthyretin is a homotetrameric protein present in serum and cerebral spinal fluid and the main function is the transport of L-thyroxine and the holo-retinol-binding-protein. The drug is used in the form of a meglumine salt and functions by kinetic stabilization of the correctly folded tetrameric form of the protein. In the pathologic state the protein dissociates under denaturing conditions and the following monomer unfolding enables amyloid fibril formation, causing finally a failure of the autonomic nervous and/or the peripheral nervous system. The treatment slows the dissociation of the native protein and consequently amyloid fibril formation and degeneration of post-mitotic tissue is delayed.

The fundamental concept for the use of Tafamidis was inter alia described in WO 2004/056315 A2. This patent document reveals that the kinetic stabilization of the native state of transthyretin is an effective mechanism for preventing protein misfolding and that inhibiting of misfolding can be used as an effective treatment or prophylaxis for such diseases. The patent document further discloses treatment and screening methods, as well as specific transthyretin stabilizing compounds.

2

In addition, a further patent document is specially directed to crystalline solid forms of 6-carboxy-2-(3,5-dichlorophenyl)-benzoxazole. WO 2016/038500 A1 is for instance related to solid forms of 6-carboxy-2-(3,5-dichlorophenyl)-benzoxazole and to methods for their preparation. The teaching also includes pharmaceutical compositions containing at least one solid form and the therapeutic or prophylactic use of such solid forms and compositions.

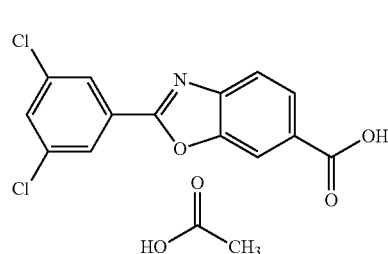
A specific crystalline form of 6-carboxy-2-(3,5-dichlorophenyl)-benzoxazole meglumine is in addition disclosed in WO 2013/038351 A1. Said crystalline form has a powder X-ray diffraction pattern comprising peaks at diffraction angles (2θ) of 10.7 ± 0.2 , 11.8 ± 0.2 , and 13.3 ± 0.2 .

Nevertheless, besides the existing routes of synthesis and the known solid forms of Tafamidis there is still the need for further routes and polymorphs, which are able to deliver high quality crystalline Tafamidis polymorphs in a reproducible and environmentally friendly way. Furthermore, there is also the need for new Tafamidis meglumine polymorphs comprising superior production properties especially with respect to handling and in tableting.

BRIEF DESCRIPTION OF THE INVENTION

Above mentioned problem is solved according to the invention by a process for the production of crystalline Tafamidis polymorphs, at least comprising the steps of:

a) Forming a dispersion by contacting a Tafamidis acetic acid adduct (E) and a solvent capable of removing the acetic acid adduct molecule from the Tafamidis;



b) Heating the dispersion obtained in step a) and

c) Precipitation and drying of the precipitate to yield the crystalline Tafamidis polymorph. Surprisingly, it has been found that it is possible to generate different crystalline Tafamidis polymorphs by starting from a Tafamidis acetic acid adduct. The exchange of the adduct molecule can be achieved in a very gently routine and the resulting products comprise a high level of crystallinity. The resulting products comprise very reproducible physical, chemical and biological properties and the amount of unwanted by-products is very low. This finding is surprising, because it cannot be expected that starting from a single adduct different crystalline polymorphs are accessible. This is surprising, because the thermodynamic and the kinetic side conditions of the polymorph forming process can be expected to be different for different adducts, usually resulting in highly specialized crystallization routines. Albeit, the chemical and sterically differences of the adduct molecules, the applicants were able to establish a very flexible process for the generation of the different adducts with advantages in terms of process chemistry as well as purity of the involved intermediates and products.

A crystalline Tafamidis polymorph according to the invention is substance at least comprising Tafamidis and